Exact, Bounded, or Relaxed: Proving Which Floating-Point Guarantee a Compiler Rewrite Keeps

ANONYMOUS AUTHOR(S)

Compilers offer a binary choice for floating-point code: strict IEEE 754 semantics, or "fast-math" permissions under which reassociation and identity rewrites may change results. The choice conflates three guarantees. A rewrite may be *bit-identical*, because every value it touches is exact or rounded once; it may change bits while keeping the source program's worst-case error bound; or it may hold only over the real numbers. We show that a compiler can prove, rewrite by rewrite, which of the three holds from facts about the values a program will see, and can certify every output with the weakest guarantee that reaches it. A grid–magnitude abstraction proves that a matrix chain whose proper sub-chains fit in the significand, and whose final sums round at most once, returns the correctly rounded product under every bracketing and order of summation; one step above this threshold, inputs exist on which bracketings differ. The standard componentwise error bound of a chain is the same for every bracketing, which licenses bound-preserving reassociation under a range condition. We implement the guarantees twice. In a matrix-language compiler with declared input domains, strict compilation runs integer kernels at 83–113% of the speed of reordering without proofs, with identical bits on every build, machine and C compiler we tried, on generated data and on the Cora citation graph and MNIST. In LLVM, a pass derives the same facts from integer conversions in ordinary C and marks a reduction reassociable only when every order is exact, statically or behind a run-time guard: Clang then vectorizes nine integer reductions 5.09–9.71× faster, as fast as `-ffast-math`, and keeps the strict bits on all 784 adversarial inputs, where `-ffast-math` and a mutant that admits one bit more do not. In four existing code bases the pass guards 23 of 109 reductions; in CPython's `audioop` module they run up to 8.05× faster without changing any of 404 results.

CCS Concepts: • **Software and its engineering** → **Compilers**; • **Theory of computation** → *Program analysis*; • **Mathematics of computing** → *Mathematical software.*

Additional Key Words and Phrases: floating-point arithmetic, compiler optimization, fast-math, reassociation, vectorization, abstract interpretation, numerical contracts, matrix chains, LLVM

1 INTRODUCTION

Floating-point addition and multiplication are not associative, and $x \cdot 0 = 0$, $x + 0 = x$ and $x \cdot I = x$ fail for signed zeros, NaN and infinities [1, 2]. Compilers therefore forbid such rewrites by default and offer permissions that enable them: GCC's `-ffast-math` and its components [3], or LLVM's per-instruction fast-math flags [4]. A permission states what the compiler *may* do. It does not state what the result *is*: after `-fassociative-math` a programmer cannot tell which outputs still equal the strict program, which changed within a known error bound, and which changed arbitrarily.

These are three different guarantees, and a single switch conflates them. Consider the chain $R = ABC$ with $A, C \in \mathbb{R}^{100\times 2}$ and $B \in \mathbb{R}^{2\times 100}$. Source order costs 69,800 scalar operations; $A(BC)$ costs 1,396. If A, B and C hold 8-bit integers, every product and partial sum of either bracketing is an integer of at most 29 bits, so both bracketings compute the same bits: the rewrite is *bit-identical*, and a strict compiler may apply it. If the matrices hold real values in $[-1, 1]$, the bits change, yet the textbook componentwise error bound of the chain is the same for every bracketing: the rewrite is *bound-preserving*. If values are unbounded, one bracketing may overflow where the other does not, and the rewrite is *relaxed*: valid over the reals and nowhere else.

The difference between these cases is not in the program text but in the values the program will see (Table 1). We make that knowledge explicit: program inputs declare a value domain (`input(int8)`, `input(real(1))`), the runtime checks each input against its domain before execution, and the compiler derives from the domains facts precise enough to decide, rewrite by rewrite, which

Table 1. One chain $R = ABC$ (40×2, 2×40, 40×2) over three input domains: modeled arithmetic after optimization (11,120 in source order) and the certificate of `print(R)`, as our compiler reports them.

Inputs	strict	bounded	algebraic
`int8`	556, bit-identical	556, bit-identical	556, bit-identical
`real(1)`	11,120, bit-identical	556, bound-pres.	556, bound-pres.
`real`	11,120, bit-identical	11,120, bit-identical	556, relaxed

guarantee holds. Each output is then certified with the weakest guarantee among the rewrites that reach it, and the same facts tell a C compiler which sums it may reorder.

The facts need not come from a new language. Ordinary C states many of them in its types, because integer data is stored as integers and converted where it is summed:

```
double sum(const uint8_t *p, long n) { double s = 0; for (long i = 0; i < n; i++) s += p[i];
    return s; }
```

Under strict semantics Clang keeps this loop scalar, because reordering a floating-point sum may change it; with `-ffast-math` it vectorizes the loop and runs it 6.57× faster on the MNIST training images. Yet every partial sum of every order is an integer below 2^{53} whenever $n \le 2^{53}/255$, so every order returns the same bits. Our LLVM pass proves this from the `uint8_t` conversion, checks n once on entry to the loop, and lets the vectorizer reorder exactly this sum: the strict program runs 6.50× faster and prints the same result.

Contributions.

- A lattice of three numerical contracts (strict, bounded, algebraic) whose proof obligations are per rewrite, and per-output certificates that compose them, with the inheritance rule that keeps certificates sound when a proof reads facts produced by a weaker rewrite, and its soundness argument (Section 3).
- A grid–magnitude abstract domain and a theorem for matrix chains: if every proper subchain fits in the significand and the final sums round at most once, every bracketing and every order of summation returns the correctly rounded product, including the sign of zero (Section 4); and the same argument for reductions with an arbitrary start value, which a run-time guard can check. The threshold is the true boundary on three of the four chain families we measured and one step from it on the fourth.
- A proof that the standard componentwise bound of a chain is invariant under bracketing, which reduces bound-preserving reassociation to a range check (Section 5), and a segmented chain-ordering algorithm that applies the cheapest bracketing each contract admits (Section 6).
- Two implementations: a compiler for a matrix language with declared input domains, whose C backend turns certificates into OpenMP reduction clauses (Section 8); and ExactReduce, an LLVM pass that derives the facts from integer conversions in C and licenses reassociation per reduction, statically or by loop versioning (Section 9).
- An evaluation of nine research questions: recovery, certificate soundness, threshold tightness, end-to-end value under GCC and Clang on generated and real data and on two machines, test adequacy by mutation, a preregistered accuracy study, and, for the LLVM pass, speed, soundness under adversarial inputs, and how often existing code has reductions it can license (Section 11).

2 SETTING

MatrixLang is a statically shaped, straight-line language over binary64 matrices and scalars. Its compiler (flex, bison, C11) performs shape checking, lowers to three-address code (TAC), optimizes, and executes on a typed stack virtual machine (VM). A matrix product entry is evaluated as $c_{ij} = (\cdots((+0 + a_{i1}b_{1j}) + a_{i2}b_{2j})\cdots) + a_{in}b_{nj}$: the accumulator starts at +0, each term is one rounded multiplication and one rounded addition in round-to-nearest, and no fused multiply-add is used (`-ffp-contract=off`). Outputs are compared in hexadecimal, which distinguishes every binary64 value including -0 and NaN.

The optimizer performs constant folding, common subexpression elimination, copy propagation and dead-code elimination, which are bit-exact; algebraic identity and zero rewrites ($x \cdot 1$, AI, $x + 0$, $A \cdot 0$, $0 \cdot A$); and matrix-chain reordering by dynamic programming over the modeled cost $m\,p\,(2n-1)$ of an $m \times n$ by $n \times p$ product. Before this work a single flag decided all value-changing rewrites at once.

Input domains. A declaration `matrix A[40,2] = input(int8);` introduces a run-time input whose values the compiler never sees. Domains are `bool`, `uint8`, `int8`, `uint16`, `int16`, `int32`, `int(lo,hi)`, `real` (any finite value), `real(B)` ($|x| \le B$) and `real(lo,hi)`. Values arrive from files or a seeded generator and are checked before execution: finite, integral for integer domains, within bounds, and with -0 stored as +0 in integer and non-negative domains. The check is what makes domain-derived facts sound; without input declarations every value would be a compile-time constant and constant folding would subsume optimization.

LLVM. LLVM attaches fast-math flags to individual floating-point instructions [4]: `reassoc` permits reassociation, `nsz` lets the sign of a zero be ignored, `nnan` and `ninf` assume the absence of NaN and infinity, and so on. The loop vectorizer recognizes an addition reduction from its header phi and vectorizes and interleaves it only when every addition of the chain carries `reassoc`, or when a loop pragma forces vectorization, itself a per-loop license [5]; otherwise a floating-point reduction stays scalar on targets without ordered vector reductions, which is why strict semantics cost speed.

3 CONTRACTS AND CERTIFICATES

Definition 1 (Guarantee levels). *A rewrite of program P into P' is* bit-identical *if every output of P' equals the corresponding output of P bit for bit, for every input in the declared domains;* bound-preserving *if every output of P' satisfies the standard-model worst-case error bound of P's evaluation order and NaN and infinity occur in P' exactly where they occur in P; and* relaxed *otherwise, when it is an identity over the reals.*

The levels are ordered bit-identical < bound-preserving < relaxed. A *contract* names the weakest level a rewrite may have and still be applied: *strict* (the default) admits only bit-identical rewrites, *bounded* admits bound-preserving ones, and *algebraic* admits all. Each candidate rewrite computes its level from the facts of its operands; the contract applies or declines it, and a declined rewrite is reported with the contract that would permit it. Constant folding, common subexpressions, copy propagation, dead-code elimination and double transposition are bit-exact and need no proof.

Certificates. Every TAC instruction carries the weakest level of the rewrites applied to it. The certificate of an output is the weakest level in the dependency cone of its `print`, together with the reasons recorded by those rewrites. The compiler prints, for example:

```
Contract: bounded. Every output must keep the source order's worst-case error bound.
 1. bit-identical     print(R)
```

```
   - A * B * C is exact: every intermediate of every bracketing needs at most 51 of 53
     significand bits
2. bound-preserving  print(S)
   - X * B * C keeps the source-order error bound: |error| <= 2.89e-15 |A1|...|Ak|
     under every bracketing, and no intermediate can overflow
```

Inheritance. A dependency cone in the *optimized* program is not enough. In $R = 0 \cdot (1 \cdot (0 \cdot N)^{\mathsf{T}})$ with $N = -A$, the inner $0 \cdot N$ may be -0; the bounded contract rewrites it to $+0$ as a bound-preserving rewrite. The outer product is then provably $+0$, a bit-identical rewrite, but only because of the inner one, and once the outer rewrite drops every use of the inner product, dead-code elimination removes it and its label. Our soundness experiment (Section 11.2) found exactly this case in an earlier version. A rewrite now inherits the weakest level in the dependency cone of the operands its proof read, at the time of the rewrite.

Proposition 3.1 (Bit-identical certificates are sound). *Let optimization apply rewrites $r_1, \ldots, r_n$ in order, each with a level computed from facts that over-approximate its operands' values in the program it rewrites, and let labels follow the inheritance rule. Then every instruction labelled bit-identical computes, for every input in the declared domains, the value of the instruction of the unoptimized program it stands for; in particular every output certified bit-identical prints the unoptimized bits.*

Proof. By induction on k, for the program after $r_1, \ldots, r_k$. Initially every instruction is its own counterpart. Bit-exact passes replace instructions by others that compute the same value from the same operands, so they preserve the claim. Let r_k be bit-identical and replace instruction t by t'. Its proof establishes $t' = t$ for every operand value that satisfies the facts it read, and the facts hold for the operands as computed in the current program, so t' computes what t computed there. If every instruction in the cone of those operands is labelled bit-identical, the operands equal their unoptimized counterparts by the induction hypothesis, hence so does t'. Otherwise the inheritance rule labels t' with the weaker level, and the claim does not apply to it. Rewrites at a weaker level label their results weaker as well. Dead-code elimination removes instructions but not labels of instructions that remain, because each label was fixed when its instruction was created. □

The certificate is a claim over the whole declared domain; executing a program checks it on drawn inputs only (Section 11.2).

4 EXACTNESS FROM INPUT DOMAINS

The grid–magnitude domain. A fact about a matrix X is a tuple $(g, m, \mathit{fin}, \mathit{nn}, \mathit{nz})$: every entry is a multiple of 2^g ($g \geq -1074$), has magnitude at most m, is finite, is non-negative, and is not -0, respectively. Integer domains give $g = 0$ and $m = \max(|lo|, |hi|)$; bounded reals give $g = -1074$ (every binary64 value is a multiple of 2^{-1074}) and $m = B$; unbounded reals give $m =$ DBL_MAX. Literals give exact facts. Transfer functions use upward-rounded bound arithmetic: a sum takes the smaller grid and adds magnitudes; a product of entries adds grids; a matrix product of inner dimension n has grid $g_a + g_b$ and magnitude $n\, m_a m_b$, inflated by $(1 + 2nu)$ when not provably exact. Non-finite or overflowing results lose every fact. A matrix-product entry is never -0, because its accumulator starts at $+0$.

Lemma 4.1 (Rounding preserves grids). *If x is a real multiple of 2^g with $g \geq -1074$ and $|x| \leq$ DBL_MAX, then $\mathrm{fl}(x)$ is a multiple of 2^g, and $\mathrm{fl}(x) = x$ when $|x| \leq 2^{g+53}$.*

Proof. Let $e = \lfloor \log_2 |x| \rfloor$; the spacing of binary64 near x is $2^{\max(e-52, -1074)}$. If $|x| < 2^{g+53}$ then $e - 52 \leq g$, so the spacing divides 2^g and x is representable. If $|x| = 2^{g+53}$, x is a power of two in

range. Otherwise $e \geq g + 53$, the spacing 2^{e-52} is a multiple of 2^g, and so is every representable neighbor of x. □

The same holds for binary32 with 24 significand bits and $g \geq -149$.

Lemma 4.2 (Signed zeros of exact sums). *In round-to-nearest, a sum of two multiples of 2^{-1074} is -0 only if both addends are -0. Hence a summation tree whose additions are all exact returns -0 if and only if every leaf is -0, whatever its shape; in particular a tree with a $+0$ leaf never returns -0.*

Proof. An exact zero sum of addends that are not both -0 is $+0$, and a non-zero multiple of 2^{-1074} never rounds to zero. By induction, a subtree is -0 exactly when every leaf is. □

Let $A_1 \cdots A_k$ be a chain with $A_l \in \mathbb{R}^{d_{l-1} \times d_l}$ and facts (g_l, m_l). For $i \leq j$ let $G_{ij} = \sum_{l=i}^{j} g_l$ and $M_{ij} = \prod_{l=i}^{j} m_l \cdot \prod_{l=i}^{j-1} d_l$, which bounds every entry of $|A_i| \cdots |A_j|$. Sub-chain $i..j$ *fits* if $M_{ij} = 0$, or $G_{ij} \geq -1074$ and $M_{ij} \leq 2^{G_{ij}+53}$.

Theorem 4.3 (Correctly rounded chains). *Let every operand be finite and every proper sub-chain $i..j \neq 1..k$ fit. Let the whole chain fit, or satisfy $G_{1k} \geq -1074$, $M_{1k} \leq$ DBL_MAX and $(d_s - 1)\, M_{1s} M_{s+1,k} \leq 2^{G_{1k}+53}$ for every split $1 \leq s < k$. Then every bracketing, with the terms of every product entry summed in any order from a $+0$ accumulator, returns $\mathrm{fl}(A_1 \cdots A_k)$ entrywise, and no entry is -0. All such evaluations are bit-identical.*

Proof. Every non-root node T of a bracketing computes its proper sub-chain $P_{ij} = A_i \cdots A_j$ exactly. By induction on $T = LR$ splitting after s: the children are exact, every term $\ell_{ab} r_{bc}$ is a multiple of $2^{G_{is}+G_{s+1,j}} = 2^{G_{ij}}$, and the sum of any subset of the d_s terms has magnitude at most $d_s M_{is} M_{s+1,j} = M_{ij} \leq 2^{G_{ij}+53}$, so by Lemma 4.1 every product and every addition, in any order, is exact. If $M_{ij} = 0$, every term is an exact zero of finite factors.

The root, split after s, has exact children. If the whole chain fits, the root is exact as well. Otherwise every term is at most $M_{1s} M_{s+1,k}$ and every sum of at most $d_s - 1$ terms at most $(d_s - 1) M_{1s} M_{s+1,k} \leq 2^{G_{1k}+53}$, so all of them are exact (when $d_s = 1$, the single product is correctly rounded by IEEE 754). In any summation tree, the lowest node whose subtree holds all d_s terms adds two exact partial sums and returns $\mathrm{fl}(c)$ for the exact entry c; any addition above it adds only the accumulator's $+0$. Lemma 4.2 excludes -0. Every bracketing and every order thus returns $\mathrm{fl}(A_1 \cdots A_k)$. □

A proper sub-chain must be exact because some bracketing uses it as an operand, and a rounded operand changes everything computed from it. Checking the whole chain alone is unsound: if the last operand is a zero matrix, $M_{1k} = 0$, yet $A_1 A_2$ may overflow to ∞ and $\infty \cdot 0 =$ NaN in source order, while the reordered chain computes 0; our mutation study contains this witness (Section 11.5). For `int(0,m)` chains of shape $100 \times 2 \cdot 2 \times 100 \cdot 100 \times 2$ the binding condition is the source order's last product, and the largest admissible bound is $m^\star = 35{,}697$, since $198 \cdot 35697^3 \leq 2^{53} < 198 \cdot 35698^3$. Requiring the whole chain to be exact stops at 35,578; Section 11.3 shows that 35,697 is the true boundary.

Corollary 4.4 (Free reductions). *A product LR with inner dimension d, finite operands, $g_L + g_R \geq -1074$ and $(d-1)\, m_L m_R \leq 2^{g_L+g_R+53}$ gives the same bits for every order of summation of every entry.*

A reduction in C need not start from $+0$, and its length may be known only at run time. The same argument covers both, and its condition can be checked on entry to a loop.

Proposition 4.5 (Guarded reductions). *Let p be the significand width and $e_{\min}$ the smallest exponent of the format, $G \geq e_{\min}$ with 2^{p+G} finite, and let $s_0, t_1, \ldots, t_N$ be finite multiples of 2^G with $|t_i| \leq K 2^G$. If $|s_0|/2^G + NK \leq 2^p$, then every evaluation of $s_0 + t_1 + \cdots + t_N$ as a summation tree,*

in any association and order, and with any number of -0 leaves added, returns the exact sum, and returns -0 only when s_0 and every t_i are -0.

Proof. Every node sums a subset of the leaves: a multiple of 2^G of magnitude at most $|s_0| + NK2^G \le 2^{p+G}$, which is representable (Lemma 4.1). So every addition is exact and every node holds the exact sum of its leaves; added -0 leaves do not change it. Lemma 4.2 gives the sign of a zero result, which depends only on the leaves. □

A vectorized reduction is such a tree: its partial sums start from the identity -0 in every lane but one, which holds s_0.

5 BOUND-PRESERVING REASSOCIATION

In the standard model $\mathrm{fl}(a \circ b) = (a \circ b)(1+\delta)$, $|\delta| \le u = 2^{-53}$, absent underflow and overflow, an inner product of length n evaluated in any order satisfies $|\mathrm{fl}(x^{\mathsf{T}}y) - x^{\mathsf{T}}y| \le \gamma_n |x|^{\mathsf{T}}|y|$ with $\gamma_n = nu/(1-nu)$ [7].

Theorem 5.1 (Bracketing invariance). *For every bracketing T of $A_1 \cdots A_k$ evaluated without underflow or overflow,*

$$|\hat{T} - A_1 \cdots A_k| \le \left(\prod_{l=1}^{k-1} (1+\gamma_{d_l}) - 1 \right) |A_1| \cdots |A_k|$$

componentwise, where $d_1, \ldots, d_{k-1}$ are the inner dimensions.

Proof. Write $\Pi_T = \prod(1+\gamma_d)$ over the dimensions contracted inside T and $|A|_T$ for the product of the operands' absolute values. We show $|\hat{T} - P_T| \le (\Pi_T - 1)|A|_T$ by induction. For $T = LR$ contracting d, let $\hat{L} = P_L + E_L$ and $\hat{R} = P_R + E_R$ with $|E_\cdot| \le (\Pi_\cdot - 1)|A|_\cdot$, so $|\hat{L}| \le \Pi_L |A|_L$. Then $\hat{T} = \hat{L}\hat{R} + F$ with $|F| \le \gamma_d |\hat{L}||\hat{R}|$ and $\hat{L}\hat{R} - P_L P_R = E_L P_R + P_L E_R + E_L E_R$. Using $|P_\cdot| \le |A|_\cdot$,

$$|\hat{T} - P_T| \le \left[(\Pi_L - 1) + (\Pi_R - 1) + (\Pi_L - 1)(\Pi_R - 1) + \gamma_d \Pi_L \Pi_R \right] |A|_T = (\Pi_L \Pi_R (1+\gamma_d) - 1)|A|_T.$$

Every inner dimension is contracted at exactly one node of any bracketing, so $\Pi_T = \prod_l (1+\gamma_{d_l})$ for every T. □

The bound coefficient depends neither on the bracketing nor, since γ_d holds for every order, on the order of summation inside each product, so a rewrite between bracketings preserves it whenever both are evaluated without overflow, and NaN and infinity cannot arise in either. The bounded contract therefore admits a reordering when every sub-chain's magnitude bound stays below 2^{1020} (leaving slack for the Π factors). This is why `real(1)` chains are reordered under the bounded contract while chains over unbounded `real` are not: there no fact excludes overflow. Theorem 5.1 also explains why an earlier attempt to trade cost against worst-case accuracy in the chain dynamic program was degenerate: all bracketings tie. Underflow adds an absolute term to the standard model; we do not claim that term is bracketing-invariant (Section 13).

6 SEGMENTED CHAIN ORDERING

A contract may admit a cheaper bracketing for part of a chain but not the whole. For a chain $A_1 \cdots A_k$ and source-order prefixes $P_i = A_1 \cdots A_i$ (computed left to right, as the source does), the pass computes

$$best[j] = \min_{i<j} \; best[i] + \mathrm{cost}(P_i \cdot A_{i+1} \cdots A_j),$$

where a segment of two operands is the source step, always legal, and a longer segment may use its cheapest bracketing (the classical $O(k^3)$ program [8]) if its level is admitted: bit-identical by

Table 2. Levels of identity and zero rewrites from operand facts. "fin" is finite, "nz" has no −0 entry, "nn" is non-negative.

Rewrite	bit-identical if	else
$x \cdot 1 \to x$ (scalar unit)	always	–
$AI, IA \to A$ (product)	fin, nz	bounded if fin
$x + 0, 0 + x \to x$	x nz	bounded
$x - 0 \to x$	x nz or zero nz	bounded
$A \cdot 0 \to 0$ (product)	other operand fin	relaxed
$0 \cdot A \to 0$ (scaling)	both nn and fin	bounded if fin

Theorem 4.3 for strict, bit-identical or range-safe for bounded, anything for algebraic. Segment facts use the facts of P_i as their first operand. Ties prefer the stronger guarantee, so an exact tree is chosen over a bound-preserving tree of equal cost. Only segment roots carry a guarantee: interior products have no counterpart in the source program, and labelling them let common subexpression elimination spread one chain's proof to an unrelated output. On a four-operand `int(0,m)` chain $24 \times 3 \cdot 3 \times 24 \cdot 24 \times 3 \cdot 3 \times 24$, the whole chain is bit-identical only for $m \le 2{,}568$, but its first three operands are up to $m = 50{,}727$, and the strict contract reorders that segment (Section 11.3). Operands built in place, such as `ones(n,1)` in $A^3\mathbf{1}$, read nothing and are moved above the chain so that its products are contiguous.

7 IEEE SIDE CONDITIONS OF IDENTITY REWRITES

Table 2 lists the conditions. A matrix product with I is not the identity in IEEE 754: each entry is $+0 + a \cdot 1 + b \cdot 0 + \cdots$, which maps -0 to $+0$ and ∞ to NaN through $\infty \cdot 0$. Adding $+0$ maps -0 to $+0$. A scaling takes the sign of each factor, so $0 \cdot (-3) = -0$. The rule $0 - x \to -x$ is absent: it maps $+0 - (+0) = +0$ to -0 and saves no arithmetic.

8 CERTIFICATES AS CODE

A guarantee is worth most when a production compiler can act on it. `matrixc --emit-c` writes the optimized program as a C11 translation unit that GCC or Clang compiles like any other. Every instruction becomes a loop nest that performs the VM's arithmetic. A product entry is summed from +0 over k ascending, written i-k-j with literal trip counts so that the C compiler may still vectorize across j, interchange loops, or run rows on separate threads: none of these changes any entry's order of summation. The program reads its inputs from a file, checks them against their domains as the VM does, and prints every output in hexadecimal.

For each product the facts decide the weakest level at which its sums may be reordered: bit-identical by Corollary 4.4; bound-preserving when no partial sum can overflow, since every order keeps the bound $\gamma_d |a|^{\mathsf{T}} |b|$; relaxed otherwise. When the contract admits that level and the output is narrow (at most four columns), the emitter writes the product as dot products under `#pragma omp simd reduction(+:acc)`. OpenMP leaves the order in which such partial results are combined unspecified [6], so the clause is a standard, portable license to reassociate exactly that sum and no other. Compiled without OpenMP, the pragmas are ignored and every sum keeps the VM's order. The emitted comment states the level and its reason:

```
/* t5 = A * t3  order of summation free,
   bit-identical: every term and partial sum is exact, so every order gives the same bits */
```

The backend is also a second implementation of the VM's arithmetic, compiled by a different compiler. On 83 programs (every example and generated programs over all domains), under each

contract, GCC's build of the emitted C printed exactly the VM's output for the same optimized program, with and without OpenMP SIMD; under the strict contract it printed the *unoptimized* program's output also with every licensed reduction split across vector lanes and across four threads (0 mismatches in all). Under the bounded and algebraic contracts, licensed reductions changed the output of 27 and 33 programs.

9 THE LICENSE IN LLVM

MatrixLang's facts come from declared domains. C has no such declarations, but its programs state many of the same facts in another way: integer data is stored in integer arrays and converted where it is summed. ExactReduce, an LLVM 18 pass plugin of 955 lines of C++, derives grid facts from those conversions and licenses the reassociation of a reduction only when it proves that every order returns the same bits. It runs at the start of the vectorizer pipeline, after the loop and scalar optimizations, so that the loop vectorizer, which runs next, reads its decisions.

Facts from the IR. A fact (G, K) about a floating-point value states that it is an integer multiple of 2^G of magnitude at most $K2^G$, the grid–magnitude domain of Section 4 with exact integer bounds. An `sitofp` or `uitofp` of an integer gives G from the integer's known trailing zero bits and K from its range, which LLVM's value tracking and scalar evolution compute (the latter through the loop's guards, so that a loop counter converted to floating point is bounded too); the conversion must itself be exact. Constants give exact facts; `fmul`, `fadd`, `fsub`, `fneg`, `fabs`, `fpext`, `fptrunc`, `select`, phis and `llvm.fmuladd` propagate them as in Section 4, and every operation inside a term must be exact. A value loaded from a floating-point array has no fact: the pass licenses nothing over real-valued data.

Reductions and the static license. The pass takes every floating-point add reduction that LLVM's recurrence analysis recognizes and walks its chain from the loop latch back to the header phi: additions, subtractions of a term from the accumulator, `llvm.fmuladd` with the accumulator as addend, and the selects and phis of conditional reductions. The walk yields W, the largest magnitude one iteration can add. If the start value s_0 has a fact, either a constant or the running sum of enclosing loops that only add this loop's result to it, and scalar evolution bounds the trip count by N, Proposition 4.5 applies when $|s_0| + NW \le 2^{p+G}$; for nested sums N is the product of the trip-count bounds. Every term of a product kernel over `int8` data with an `int` trip count, for instance, is at most 2^{14} and there are at most 2^{31} of them, so every order is exact in binary64.

The guarded license. When the start value is unknown, or the trip count may be too large, as with any `long` or `size_t` loop bound, the pass versions the loop. The preheader computes

```
a  = fabs(s0);  u = a * 0x1p-G;                  /* exact: G <= 0, scaling by a power of two */
ok = a <= 0x1p(p+G) && u == trunc(u)             /* s0 is a multiple of 2^G, |s0| <= 2^(p+G) */
     && btc < (2^p - (uint64_t)u) / K;           /* |s0| / 2^G + trip count * K <= 2^p       */
if (ok) /* licensed copy of the loop */ else /* the loop as it was */
```

where `btc` is the backedge-taken count. When scalar evolution can express that count only through a division by a loop-invariant value, which it will not compute before the loop because the divisor might be zero, the guard uses the dividend, a valid upper bound. Whenever the guard holds, the condition of Proposition 4.5 holds for this entry into the loop, so the licensed copy prints the bits the original would have. A loop with several guarded reductions gets one copy, guarded by all their conditions. The guard costs a handful of instructions per entry.

What the pass changes. The pass sets `reassoc` on the reduction's additions and on nothing else. It never sets `nsz`, so the vectorizer keeps -0 as the identity of the partial sums and the trees it builds

are those of Proposition 4.5; terms and every other instruction keep their flags. The flag permits reassociation of the chain; the vectorizer uses it to split the sum across vector lanes and interleaved accumulators and to combine them at the end. A transformation that read the flag to do something other than reorder the same terms would fall outside the proposition; the stress study (RQ8) found no output that changed.

Limits. The pass needs facts: a sum of `double` array elements that happen to hold integers looks like any other, and only a declaration such as MatrixLang's could change that. It needs a bound on the trip count, or a loop it can version: innermost, with one exit and a trip count computable on entry. And a license is necessary but not sufficient: the vectorizer must also be able to vectorize the loop (RQ9 shows a loop whose run-time stride prevents it). The study's mutants are two switches of the pass: one adds a bit to every significand width, the other licenses every add reduction without a proof, which is what `-fassociative-math` does to such a loop.

10 IMPLEMENTATION

The MatrixLang compiler is 7,700 lines of C11, flex and bison. The fact domain and the segmented chain pass take 1,343 of them (`facts.c`, `chain.c`); input domains and their load-time checks take 555 (`domain.c`, `inputs.c`); contracts, side conditions and certificates are part of `optimize.c`, and the C backend is `emit_c.c`. The command line selects `--fp-strict` (default), `--fp-bounded` or `--fp-algebraic`; `--no-proofs` disables facts and reproduces the previous strict optimizer; `--certificate` prints per-output guarantees; `--random-inputs` *seed* draws inputs reproducibly; and `--emit-c` writes C. A browser workspace runs the same compiler locally, shows the three contracts and each output's certificate, checks every certificate against an unoptimized run, and presents the measurements of Section 11. ExactReduce builds against an installed LLVM 18 and is loaded with `clang -fpass-plugin=ExactReduce.so`; it reports one line per reduction with its decision and reason.

11 EVALUATION

We ask: (RQ1) how much of the algebraic optimization each contract recovers with proofs; (RQ2) whether certificates are sound; (RQ3) how close the threshold of Theorem 4.3 is to the true boundary; (RQ4) what certificates are worth to a production compiler, on generated and real data, on two machines and under two C compilers; (RQ5) which tests notice a missing side condition; (RQ6) whether cheaper bracketings are more accurate; and, for the LLVM pass, (RQ7) what its licenses are worth, (RQ8) whether they are sound under inputs built to break them, and (RQ9) how often existing code has reductions they apply to. All scripts, seeds and raw observations are in the artifact; every number below is generated from them.

Machines. RQ1–RQ6 and the first half of RQ4 ran on machine M1, an Intel Core Ultra 9 275HX (24 cores, AVX2) under Windows 11 with GCC 15.2.0. The rest of RQ4 and RQ7–RQ9 ran on machine M2, an Intel Xeon Processor 2.80GHz cloud instance (4 cores, AVX-512) under Linux with GCC 13.3.0 and Clang 18.1.3.

Workloads. RQ1, RQ2 and RQ5 use generated straight-line programs whose matrices are inputs over one value domain (`bool` to `int32`, `real(1)`, unbounded `real`, or a per-matrix mixture), with six shape profiles (heterogeneous, square, narrow, balanced, elementwise, and an algebraic profile that adds identity, zero, negation and negative-scaling expressions). Two literal populations reproduce an earlier study: small integers ("dyadic") and values spread over $10^{-12}..10^{12}$ ("broad"). RQ4 uses six kernels written as the mathematics is written, on generated data, and four on real data: the Cora citation graph [40] (2,708 papers, 5,278 undirected citations, from the copy distributed with [41])

Table 3. RQ1. Modeled arithmetic removed under each contract (pooled over 15,000 programs, ten seeds, six profiles), and recovery: the share of the algebraic contract's saving obtained over the unproved strict optimizer.

Domain	Modeled arithmetic removed (%)				Recovery (%)	
	no proofs	strict	bounded	algebraic	strict	bounded
`bool`	0.4	22.8	23.1	23.1	98.8	100.0
`uint8`	0.5	14.0	17.9	17.9	77.8	100.0
`int8`	0.5	16.5	19.6	19.6	83.6	100.0
`int16`	0.3	7.8	22.3	22.3	34.4	100.0
`int32`	0.3	7.1	19.4	19.4	35.8	100.0
`real(1)`	1.4	7.5	25.1	25.1	25.9	100.0
`real`	0.2	2.5	6.0	27.2	8.4	21.2
`mixed`	0.4	6.4	16.3	16.3	37.9	100.0
`dyadic literals`	0.1	20.2	22.4	22.4	90.2	100.0
`broad literals`	0.2	5.7	20.2	20.2	27.5	100.0

and the MNIST digits [42], with an `int8` softmax classifier we trained on its training images (92.22% test accuracy after quantization, 92.19% before). RQ7 and RQ8 use the same data in C kernels; RQ9 uses existing code.

11.1 RQ1: Recovery

Table 3 and Figure 1 report 60,000 compilations with no failures. Without proofs, strict optimization removes almost nothing. With proofs it recovers 98.8% of the algebraic saving on `bool`, 83.6% on `int8` and 77.8% on `uint8` while keeping every output bit-identical; on `int16` and `int32` the 53-bit budget is exhausted by longer chains and recovery falls to 34.4% and 35.8%. The bounded contract recovers the entire saving on every domain with a magnitude bound, and 21.2% on unbounded `real`, where it cannot exclude overflow. On the earlier study's dyadic literals the strict contract recovers 90.2%: those programs were exact all along, which explains why an earlier algebraic mode never changed their output. How far the proofs reach depends on the domain's width. Recompiling 200 heterogeneous and narrow programs with every matrix over $\texttt{int}(-(2^b-1), 2^b-1)$, the strict contract proves bit-identical 99% of the chain reorderings the algebraic contract applies at $b = 3$, 89% at $b = 7$, 46% at $b = 15$, 3% at $b = 17$, and none from $b = 27$: a product of three such operands already needs more than 53 bits. The proofs are cheap: compiled one at a time, a sample of 300 of these programs took a median 8.5 ms end to end under strict against 8.5 ms without proofs (median ratio 1.00, largest 1.11), a time dominated by process start-up.

11.2 RQ2: Soundness of certificates

Each program ran unoptimized and under every contract on three seeded input draws, 57,600 executions in all (Table 4). Strict output equalled unoptimized output on all 57,600 outputs. Under the bounded and algebraic contracts, 0 and 0 outputs certified bit-identical differed; the outputs that did differ were all certified bound-preserving or relaxed. This experiment is what found the inheritance defect of Section 3: before the fix, 12 outputs per contract, from four programs of the algebraic profile, were certified bit-identical and differed. Execution checks a certificate for drawn inputs only; the certificate itself is a claim over the whole domain, which Theorem 4.3 and Proposition 3.1 support. Certificates are also conservative: of 3,995 outputs the bounded contract certified below

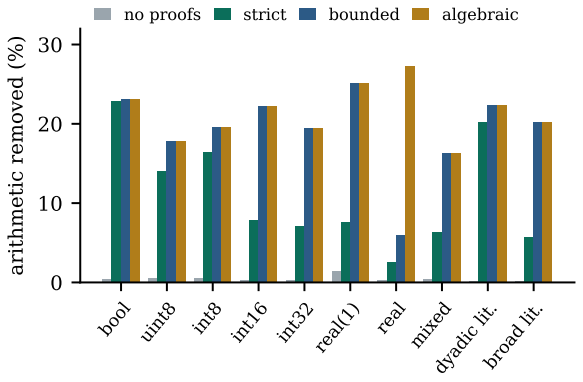


Fig. 1. RQ1. Modeled arithmetic removed per value domain and contract.

Table 4. RQ2. Outputs of 4,800 executed programs on three input seeds, compared bit for bit with the unoptimized program.

Contract	Outputs	Differ	Bit-identical		Bound-preserving	
			certified	differ	certified	differ
strict	57,600	0	57,600	0	0	0
bounded	57,600	9,202	45,615	0	11,985	9,202
algebraic	57,600	10,816	43,827	0	11,877	9,202

Table 5. RQ3. Largest m for which strict reorders `int(0,m)` chains: under the exact-only condition, by Theorem 4.3 in closed form, and found by binary search on the compiler's certificates; the smallest m with a confirmed witness; strict runs that differed from unoptimized output.

Chain	exact only	Thm. 4.3	compiler	witness	differ
100×2	35,578	35,697	35,697	$m^\star$+1	0
50×4	35,578	35,818	35,818	$m^\star$+1	0
40×8	30,419	30,676	30,676	$m^\star$+1	0
24×3×4	50,013	50,727	50,727	$m^\star$+2	0

bit-identical, 870 (21.8%) printed the unoptimized bits on all three draws; the certificate bounds what may change, not what will.

11.3 RQ3: Tightness of the threshold

For four chain families over `int(0,m)` we found the largest m for which strict reorders by binary search on the compiler's certificates (Table 5). It equals the closed form of Theorem 4.3 in every family; for the four-operand chain it is the threshold of the first segment. Allowing one final rounding raises every threshold over the exact-only condition, by 0.3–1.4%. We then looked for witnesses at 12 values of m per family, from $m^\star - 3$ to $2m^\star$: inputs on which the refused bracketing, written out explicitly and run unoptimized, prints different bits from the source order. Besides generic inputs (all-maximum, six near-maximum and ten seeded draws), a search emulates the VM's

Table 6. RQ4 on M1. Six kernels compiled by `matrixc` to C and by GCC 15.2.0 (`-O3 -march=native -ffp-contract=off`): median time of the program as written, and speedups over it of the same C under `-ffast-math`; of the strict contract; of strict with licensed reductions (`-fopenmp-simd`); of the bounded and the algebraic contracts with licensed reductions; and of strict with licensed reductions on eight threads. The algebraic contract reorders without proofs. † marks outputs that differ from the source program. Results counts distinct outputs over eleven builds, including SSE2-only builds.

Kernel	Inputs	Source	Speedup over source						Results
		(ms)	fast-math	strict	+ clauses	bounded	algebraic	8 threads	
`walks`	`bool`	342	1.12	278	288	288	284	1,202	1
`layers`	`int8, uint8`	1.14	0.99	4.98	5.47	5.44	5.69	14.6	1
`image`	`uint8`	4.61	1.02	0.98	1.02	1.05	1.05	1.87	1
`normal`	`int8, int16`	0.0288	1.09	1.27	1.28	1.24	1.25	1.31	1
`lowrank`	`real(1)`	11.8	1.01	1.00	0.93	469†	503†	6.51	2
`diffusion`	`real(0,1)`	325	1.13	1.06	1.05	277†	277†	5.06	2

arithmetic on one output entry and tries inputs at the maximum with a few entries lowered, so that a partial sum before the last one is odd and above 2^{53} and therefore rounds; every witness it finds is confirmed in the VM. No input at or below a threshold separated the bracketings (0 witnesses), and none of the 847 strict runs differed from unoptimized output. For the three three-operand families a witness exists at $m^\star + 1$: the threshold is the true boundary. For the four-operand chain the first witness is at $m^\star + 2$, and none can exist at $m^\star + 1$: that maximum is even, so an odd partial sum needs a lowered entry in the first operand's row, which brings the sum below 2^{53}. Generic inputs alone found their first witness no lower than $1.100\, m^\star$, which is why an earlier version of this study, which lacked the search, reported up to ten percent of slack.

11.4 RQ4: Value in a production compiler

Table 6 compiles six kernels written as one would write the mathematics: walk counts $A^3\mathbf{1}$ in a 1,024-vertex graph (`bool` adjacency), two stacked quantized layers W_2W_1x, row and column profiles of a $2{,}048^2$ 8-bit image, the normal equations $X^\mathsf{T}X$, $X^\mathsf{T}y$ of an integer-coded regression, and, as real-valued controls, a rank-16 update BAx and three diffusion steps P^3v. Times are medians of 50 runs in five processes. The source build printed the VM's unoptimized output on every kernel.

The large factors come from reordering chains, which changes the amount of work; any chain optimizer that ignores floating point gets them, as the algebraic column shows. What the contracts add is that a strict compiler may take them. Fast-math gains at most 1.13×: GCC reassociates scalar sums but does not reorder matrix chains. The strict contract proves the reorderings of `walks` and `layers` bit-identical and gains 278× and 4.98× with identical bits; with licensed reductions, which the algebraic build also has, 288× and 5.47×, the speed of the unproved reordering (284× and 5.69×). It declines the real-valued chains, which the bounded contract reorders within the source order's error bound for 469× and 277×. Licensed reductions add little on these kernels (0.93–1.10× over strict on one thread). The products they apply to are matrix–vector products, which use each matrix entry once, so data movement rather than the order of additions bounds their speed. Every integer kernel printed one result across all eleven builds, from SSE2 to eight threads, as Theorem 4.3 guarantees. Each real-valued kernel printed two: the source order's, and the reordered chain's, which the bounded and algebraic builds shared on this machine although nothing guarantees it.

Real data. Table 7 repeats the measurement with real inputs: walks of length three on Cora ($A^3\mathbf{1}$, `bool`), three steps of the random walk on Cora (P^3v with $P = D^{-1}A$, `real(0,1)`), the normal

Table 7. RQ4 on real data, on M2. Four programs over the Cora citation graph and MNIST, columns as in Table 6 (*threads*: strict with licensed reductions on 4 threads), compiled by GCC 13.3.0 and Clang 18.1.3. Results counts distinct outputs over all builds of both compilers.

Kernel	Build	Source	Speedup over source						Results
		(ms)	fast-math	strict	+ clauses	bounded	algebraic	threads	
`cora_walks`	GCC	42,261	0.98	1,168	1,888	2,059	1,816	6,833	1
	Clang	37,463	0.96	946	1,691	1,678	1,837	8,546	
`cora_diffusion`	GCC	41,762	1.00	0.99	0.98	2,130†	2,080†	4.28	6
	Clang	38,707	1.01†	0.95	0.94	1,532†	1,870†	3.97	
`mnist_normal`	GCC	14.8	0.98	1.02	1.10	1.07	0.98	3.30	1
	Clang	16.6	1.11	1.17	1.15	1.19	1.17	2.74	
`mnist_scores`	GCC	25.9	0.87	5.00	9.73	9.34	11.5	61.5	1
	Clang	25.1	1.13	4.86	10.2	8.91	8.74	19.7	

equations of a least-squares fit of the digit label on the central 10×10 pixels of 4,096 MNIST images (`uint8`), and the class scores of the `int8` classifier summed over 4,096 test images ($WX\mathbf{1}$, `int8`). The input checks accepted every real input in its declared domain, and with both compilers the source build printed the VM's unoptimized output for every program. The strict contract proves the walk count bit-identical on the real graph and gains 1,168× under GCC and 946× under Clang, and with licensed reductions 1,888× and 1,691×, against 1,816× and 1,837× for the unproved reordering; the classifier's scores gain 9.73× and 10.2×. Each integer program printed one result across all 22 builds. The random walk is real-valued: strict declines it (0.99× and 0.95×), and the bounded contract reorders it for 2,130× and 1,532× within the source order's error bound, printing a different result. The normal equations contain no chain, and only their narrow product $X^{\mathsf{T}}y$ takes reduction clauses, so no contract changes much.

A second machine and compiler. Table 8 repeats Table 6 on M2, with GCC and with Clang. The source program printed the same bits with all 3 toolchains for 6 of six kernels. Every integer kernel printed a single result across every build on every machine (4 of 4): the certified outputs are reproducible across vector widths, thread counts, compilers and machines. The real-valued kernels printed up to 7 results each. With licensed reductions, strict compilation ran the integer kernels at 83–113% (median 99%) of the speed of reordering without proofs, on both machines and with both compilers. On M2, whose AVX-512 units make reductions cheap once they may be reordered, the clauses matter more than on M1: the walk count gains 190× under strict GCC and 563× with them.

Hand-written reductions. The license pays where a reduction is the only parallelism. We compiled six hand-written reductions twice with GCC: strict, and with `-fassociative-math`, `-fno-signed-zeros` and `-fno-trapping-math`, which let GCC reassociate and vectorize them (Table 9). On certified data both builds printed identical bits, and the licensed build was faster by 1.23× (an `int16` dot product, memory-bound) to 6.54× (an `int8` dot product in binary32, compute-bound); on M2 with GCC 13.3.0, by 1.67–4.89×, with identical bits on all 5 certified kernels. The real-valued control changed bits for a comparable speedup. A run-time domain check over both inputs of the `int16` dot product costs 95.1% of the strict kernel; MatrixLang performs it once at load time, and ExactReduce's guard checks only the start value and the trip count. With OpenMP, whose reduction combines partial sums in an unspecified order, the certified dot product

Table 8. RQ4 on M2: the kernels of Table 6 compiled by GCC 13.3.0 and Clang 18.1.3 (*threads*: 4). Results counts distinct outputs over every build of both compilers on M2 and of Table 6 on M1.

Kernel	Build	Source	Speedup over source						Results
		(ms)	fast-math	strict	+ clauses	bounded	algebraic	threads	
`walks`	GCC	758	0.99	190	563	592	679	2,450	1
	Clang	648	1.04	159	711	751	699	1,646	
`layers`	GCC	2.8	0.88	4.31	15.6	15.5	16.1	46.5	1
	Clang	2.5	1.01	3.73	16.1	16.2	16.7	49.7	
`image`	GCC	10.7	1.01	1.09	1.54	1.57	1.56	2.40	1
	Clang	9.89	2.22	1.02	1.55	1.48	1.81	2.74	
`normal`	GCC	0.0599	0.82	1.24	1.57	1.50	1.57	1.35	1
	Clang	0.0561	1.43	1.15	1.89	1.90	1.68	1.89	
`lowrank`	GCC	29.1	0.98	0.92	0.93	1,623†	1,573†	3.14	7
	Clang	22.6	1.20†	1.04	1.01	327†	327†	3.39	
`diffusion`	GCC	736	0.99	0.97	0.99	650†	673†	2.89	7
	Clang	661	1.08†	1.03	0.98	760†	680†	3.33	

Table 9. RQ4 on M1, hand-written reductions on integer-valued binary64 (binary32 for `fdot`), sized so every partial sum is exact, compiled by GCC 15.2.0 `-O3 -march=native` strict and with reassociation licensed. Medians of 7 processes × 21 repetitions, one thread.

Kernel	strict (μs)	licensed (μs)	speedup	same bits
`dot_i16`	414.6	337.8	1.23×	yes
`sum_i32`	830.9	337.8	2.46×	yes
`matvec_u8_i16`	2,504.4	1,460.3	1.71×	yes
`gemm_i8`	4,095.1	1,259.3	3.25×	yes
`fdot_i8`	91.6	14.0	6.54×	yes
`dot_real`	412.4	341.5	1.21×	no

gave one result for every thread count from 1 to 24 and every repetition, while the real-valued one gave thirteen distinct results.

11.5 RQ5: Test adequacy by mutation

Ten mutants each drop one side condition (Table 10). Small-integer programs, the population an earlier differential study relied on, kill no test of mutant 1, which reorders without any proof: by Theorem 4.3 those programs are bit-identical under every bracketing, so the population is structurally blind to the defect. Broad-valued literals kill it in most programs. Side conditions on -0 and non-finite values are rarely exercised by random programs; one witness program per condition, written from Table 2, kills mutants 1, 2, 4, 6, 7, 8 and 9. Mutant 5 is equivalent: a non-finite fact never carries "no -0", so the -0 condition already blocks every rewrite it would admit. Mutant 2, which checks only the whole chain, and mutant 10, which lets every sub-chain round once instead of only the whole chain, are each killed by a few random programs (2 broad-valued and 4 domain tests for mutant 2, 2 and 4 for mutant 10). Mutant 3, which accepts 54 bits, survives every population whose inputs are drawn at random or at the domain's bounds, all-maximum inputs included. Only the

Table 10. RQ5. Tests that kill each mutant (number of tests per population in parentheses). A test kills a mutant when the mutant's strict-optimized output differs from the correct unoptimized output.

	Mutant	dyadic (90)	broad (90)	domain (720)	extreme (720)	fixtures (15)	witness (7)	boundary (26)
1	reorder without proof	0	51	247	96	2	2	26
2	whole chain only	0	2	4	0	0	1	0
3	54-bit threshold	0	0	0	0	0	0	24
4	AI: ignore -0	3	0	9	22	0	1	0
5	AI: ignore non-finite	0	0	0	0	0	0	0
6	$A+0$: ignore -0	1	1	6	21	1	2	0
7	$A0$: ignore non-finite	0	0	0	3	0	2	0
8	$0A$: ignore sign	4	7	45	47	1	1	0
9	negation yields no -0	4	1	15	20	0	3	0
10	one rounding in every sub-chain	0	2	4	0	0	0	0

Table 11. RQ6. Paired comparison of the cost-optimal and source bracketings on real-valued chains, and the equal-cost ablation. Sign tests are two-sided and exact; Holm-adjusted over the three confirmatory tests; median ratio of errors (cost-optimal / source) with seed-cluster bootstrap 95% intervals.

Population	chains	unequal	cheaper better	p (Holm)	median ratio [95% CI]
heterogeneous	648	635	411	3.1×10^{-13}	0.892 [0.863, 0.913]
narrow	606	587	317	0.058	0.979 [0.954, 1.000]
pooled	1,254	1,222	728	4.6×10^{-11}	0.941 [0.925, 0.955]
Equal-cost ablation: first = fewer intermediate entries, or enumeration order					
heterogeneous	64	59	27	0.6	
narrow	363	352	199	0.016	
square	900	885	426	0.28	
pooled	1,327	1,296	652	0.85	

boundary population, the witnesses of RQ3's search placed just above each threshold, kills it (24 of 26 tests). A test suite for a numerical optimizer needs inputs constructed from its proof obligations.

11.6 RQ6: Accuracy of cheaper bracketings

Theorem 5.1 says the cheaper bracketing has the same worst-case bound; it says nothing about typical error. We preregistered a study before running it: chains of three to five real-valued operands with dimensions up to 200, values uniform in $[-1, 1]$, uniform in $[0, 1]$, or broad; ten new seeds; error $\max_{ij} |\hat{c}_{ij} - c_{ij}| / (|A_1| \cdots |A_k|)_{ij} / u$ against an exact integer reference; and a numpy emulation of the VM's evaluation order, which matched `matrixc` bit for bit on 120 of 120 validation bracketings. The prediction was that the pooled sign test rejects equality at 0.01 in favour of the cost-optimal order with the same direction in every profile, and that an equal-cost ablation removes the effect.

The cost-optimal bracketing was more accurate in 728 of 1,222 chains with unequal error (Holm $p = 4.6 \times 10^{-11}$; median ratio 0.941, 95% CI [0.925, 0.955]), significantly on heterogeneous shapes (411 of 635, ratio 0.892) and in the same direction but not significantly on narrow shapes (317 of 587, $p = 0.058$), where cheaper bracketings save less. Pairs of distinct bracketings with equal modeled cost showed no difference (652 of 1,296, $p = 0.85$), as an operation-count account predicts. Because equal-cost pairs almost always also have equal intermediate sizes, the ablation does not separate that account from the rival that smaller intermediates cause the effect; we report the association, not the mechanism. No chain exceeded the bound of Theorem 5.1; the largest error reached 0.491 of it.

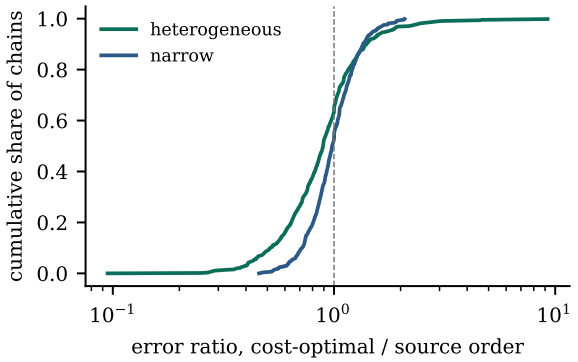


Fig. 2. RQ6. Distribution of the error ratio (cost-optimal / source) over chains whose cheaper bracketing differs from source order.

Table 12. RQ7 on M2. C reductions over integer data, compiled with `-O3 -march=native`: the pass's decision, the median time under strict Clang 18.1.3, and speedups over it of strict Clang with ExactReduce, of Clang `-ffast-math`, of ExactReduce for baseline x86-64 (SSE2), and of GCC 13.3.0 strict and `-ffast-math`. Medians of five processes of eleven runs. † marks outputs that differ from strict Clang's. Below the line, the two controls.

Kernel	Data	License	Strict	Speedup over strict Clang				
			(ms)	licensed	fast-math	lic. SSE2	GCC	GCC f.-m.
sum	MNIST	guarded	59.1	6.50	6.57	2.91	0.99	4.53
sum of squares	MNIST	guarded	59.4	6.31	6.21	2.57	0.96	4.63
Gram matrix	MNIST	static	386	5.42	5.33	1.51	0.98	3.08
int8 classifier	MNIST	guarded	93.9	9.71	10.2	3.98	0.97	6.71
degrees	Cora	static	9.11	9.19	9.25	2.91	1.01	5.95
2-walks	Cora	guarded	9.31	5.69	5.77	1.82	0.99	3.12
GEMM	generated	static	69.3	5.09	5.35	1.35	1.00	3.02
energy	generated	guarded	5.33	9.15	9.39	3.44	0.98	6.45
sum	generated	guarded	2.67	8.36	8.75	3.81	0.99	6.43
16-term sums	MNIST	guarded	23.4	1.65	1.68	0.93	0.66	1.32
dot product	generated	declined	3.81	0.97	1.54†	1.04†	1.01†	1.43†
sum, binary32	MNIST	guarded	59.9	1.02	8.48†	1.00	1.01	7.32†

11.7 RQ7: The license in LLVM

Table 12 compiles eleven kernels written as C programs write them, with integers in integer arrays: the sum and the sum of squares of the `uint8` MNIST training pixels (47 million terms), the Gram matrix of the central 10×10 pixels over all 60,000 images, the `int8` classifier's scores for the 10,000 test images accumulated in binary32, vertex degrees and walks of length two on Cora, an `int8` GEMM, the energy of `int16` samples, a sum of `int32` values, and two controls: a real-valued dot product, and a binary32 sum of the MNIST pixels, which is exact only for at most 65,793 terms. ExactReduce licensed 3 reductions statically and 7 behind a guard, and declined the real-valued one.

Table 13. RQ8 on M2. Stress cases whose output differs from strict Clang's, per build: 784 cases over 11 kernels, 7 input families and trip counts up to twice each guard's limit, split into integer kernels and the real-valued control. Strict builds disagree with each other on the control because Clang contracts $s + ab$ into a fused multiply-add where the target has one (its default `-ffp-contract=on`).

Build	integer (707)	real (77)
Clang + ExactReduce	0	0
Clang + ExactReduce, SSE2	0	35
GCC, strict	0	35
Clang `-ffast-math`	14	22
GCC `-ffast-math`	14	42
mutant: one bit more	10	0
mutant: no proof	14	22

With the license, strict Clang ran the nine integer kernels 5.09–9.71× faster, between 95% and 102% of the speedup of `-ffast-math`, and every one printed the strict bits; so did `-ffast-math` on these kernels, because their sums are exact in every order. The difference is that `-ffast-math` reassociates without knowing that: on the real-valued control it changed the result (1.54×), and on the binary32 sum, whose guard sends the 47 million terms to the strict loop, it ran 8.48× faster and printed a different sum, while the licensed build ran at the strict speed (1.02×) and printed the strict bits. With SSE2 only, the license still gained 1.35–3.98×. GCC's strict builds ran at Clang's strict speed, and GCC's `-ffast-math` 3.02–6.71×. To expose the guard's cost, the row *16-term sums* calls the sum kernel once for every run of 16 consecutive pixels, so that every call evaluates a guard for a loop of 16 iterations: the licensed build ran at 1.65× the strict speed, and `-ffast-math` at 1.68×. Compiling the kernels took a median 178 ms with the pass against 138 ms without (29% more); loop versioning grew their code by 40%.

11.8 RQ8: Soundness of the LLVM license

Random inputs rarely break an unsound reassociation (RQ5): sums of random integers stay far below 2^{53}. The stress harness therefore runs every kernel on families built from the proof obligation: all-maximum and all-minimum values, alternating signs that cancel, products that are -0, sparse maxima, and odd maxima, whose partial sums are odd and round as soon as they exceed the significand; and at trip counts that include each guard's limit, one past it, and twice it. Table 13 compares every build with strict Clang. The licensed build printed the strict bits in all 784 cases, with AVX-512 and with SSE2 alike on the integer kernels. `-ffast-math` changed 14 integer cases, all at trip counts past a guard's limit, where the strict order rounds (0 at or below one). The mutant that admits one more significand bit changed 10, all past the true limit and at most twice it (0 at or below it): the condition of Proposition 4.5, like the threshold of RQ3, is not conservative but necessary. The mutant that licenses without proof behaved as `-ffast-math`.

11.9 RQ9: Existing code

Table 14 compiles the C sources of CPython's `audioop` module, which Python 3.13 removed from the standard library [43] and the `audioop-lts` package carries; Pillow's imaging core; SciPy's `ndimage` [44]; and the 30 kernels of PolyBench/C 4.2.1 [45]. Of 109 reductions, the pass licensed 23 behind a guard and none statically: in every case a run-time length or start value had to be checked. All are sums of integer samples or pixels: `audioop`'s average, root mean square, peak-to-peak average and the 16-bit correlation behind `findfit`, `findmax` and `findfactor`, and Pillow's

Table 14. RQ9 on M2. Floating-point add reductions that LLVM recognizes in four existing code bases, compiled with `clang -O3 -march=native` and ExactReduce, and why the declined ones were declined: no grid fact (real-valued data), a term or sum that is never exact, or a loop the pass cannot version.

Code base	Files	Add reductions			Declined: why		
		static	guarded	declined	no fact	inexact	shape
CPython `audioop`	1	0	16	2	0	1	1
Pillow `libImaging`	68	0	7	15	15	0	0
SciPy `ndimage`	7 of 9	0	0	39	39	0	0
PolyBench/C	30	0	0	30	30	0	0

reduction of 32-bit images. It declined 84 reductions over real-valued data, including all 30 in PolyBench, 1 whose terms are never exact (the root mean square of 32-bit samples squares values of up to 2^{31}), and 1 in loops it cannot version. Compiling the 106 files that build took 27.9 s with the pass against 27.8 s without (best of three runs per file).

To check the licenses end to end, we compiled `audioop` with and without the pass, loaded each build in its own interpreter, and called every function with a reduction on fragments of every sample width, lengths from 0 to $2^{22} + 1$ samples (on both sides of the 16-bit guard's limit) and the families of RQ8: 404 of 404 results were identical. `findfit` on a 2,000-sample reference ran 5.78× faster; `avg` and `rms` of 8-bit samples 8.05× and 6.55×. For 16-bit samples the same two functions gained nothing (0.96×, 1.05×): their loops advance by the run-time sample width, and the vectorizer guards its vector loop with the speculation that this stride is one byte, so wider samples run the scalar loop. Existing code thus contains reductions that the license makes faster without changing a result, but fewer than kernels written for the study: most floating-point reductions in scientific code sum floating-point data.

11.10 Threats to validity

Internal. Every certificate and license was checked against execution on drawn and constructed inputs only; the theorems carry the claim over the whole domain, and RQ3, RQ5 and RQ8 test the boundaries where a wrong condition would show. The two implementations share no code but share their authors' reading of IEEE 754; the C backend checks the VM's arithmetic with a second compiler, and the LLVM study compares against Clang's own strict build. *External.* The MatrixLang workloads are generated or written by us; the real data are two public data sets, and RQ9's code bases were chosen because they compile without their build systems, not at random. RQ9 counts reductions LLVM recognizes after inlining, which is not the number a programmer would count. *Construct.* Modeled arithmetic is a cost model, not runtime; runtime appears in RQ4, RQ7 and RQ9 on two machines, one of them a shared cloud instance whose timing noise we reduce by medians over processes, not eliminate.

12 RELATED WORK

Fast-math semantics. GCC's `-fassociative-math` requires `-fno-signed-zeros` and `-fno-trapping-math` [3], and LLVM attaches `reassoc`, `nsz`, `nnan` and other permissions to individual instructions [4]. Permissions are finer than a global switch but still state what may change, not what holds. Icing gives fast-math style optimizations a semantics in a verified compiler [9], and RealCake verifies such optimizations against a real-valued semantics with a proved error bound [10]; there the program opts in and the bound is the guarantee. Our levels are derived rather than granted, so a strict program gets reassociation exactly where it is

provably bit-identical, and ExactReduce grants LLVM's own permission only where it is. Alive and Alive-FP verify peephole optimizations, including floating-point preconditions on signed zeros and non-finite values [11, 12]; Alive-Infer infers such preconditions [13]; Alive2 validates LLVM translations [14]. Our side conditions (Table 2) are preconditions of this kind; our contribution is their organisation into guarantee levels, their discharge from domain facts, and per-output certificates. Boldo et al. verify that CompCert preserves floating-point semantics exactly [15]; Monniaux catalogues why compiled floating-point code is hard to reason about [16], among them that reassociation breaks compensated summation [17], which our licenses never touch because such sums are not exact.

Value facts in production compilers. GCC's ranger tracks floating-point ranges together with whether a value may be NaN and the sign of its zeros [18], and LLVM's `KnownFPClass` analysis, since LLVM 17, tracks whether a value may be NaN, infinite or -0 [19]. Both can discharge side conditions like those of Table 2 for single instructions. LLVM also folds floating-point arithmetic on integer-to-float conversions into integer arithmetic when both are exact [20], a one-operation instance of our exactness argument. None of these tracks the grid of a value, proves a reassociation bit-identical, or certifies outputs. LLVM's loop vectorizer can version loops on run-time conditions such as pointer aliasing and unit strides [5]; ExactReduce versions on the exactness condition of Proposition 4.5.

Accuracy analysis and repair. Rosa and Daisy compile real-valued specifications to finite precision under error bounds [21, 22]; FPTaylor bounds round-off of expressions [23]; Herbie rewrites expressions for accuracy [24]; Precimonious lowers precision where a test still passes [25]. CIGEN searches for input ranges on which optimization levels make results differ [26]; we prove, for declared ranges, which rewrites cannot.

Exactness and reproducibility. The Ozaki scheme splits floating-point matrices into slices whose products are computed without error [27] and emulates binary64 GEMM on integer units [28, 29]; FFLAS-FFPACK bounds intermediates below 2^{53} to do exact integer linear algebra on floating-point units [30]. These libraries rest on the arithmetic fact that Theorem 4.3 turns into a compiler legality criterion over chains of any bracketing, with signed zeros, sub-chains and one final rounding. Error-free transformations make sums exact or accurate [31]; reproducible summation makes results independent of order at a cost [32, 33]; Intel's oneMKL reproduces results run to run for a fixed thread count and code path [34]; FPRev shows that libraries' accumulation orders differ [35]. Where our certificates apply, reproducibility across thread counts, vector widths, compilers and machines costs nothing (RQ4, RQ8). Bitwidth analysis computes value ranges to size hardware [36]. The bracketing invariance of Theorem 5.1 follows from the standard analysis of inner products [7]; we claim its use as a contract, not the inequality.

Matrix chains and testing. Optimal chain ordering is classical [8]; Linnea generates linear-algebra code with property-aware cost models [37]. Neither certifies numerical guarantees per output. Csmith-style differential testing [38] and mutation analysis [39] are established; RQ5 and RQ8 show a structural reason why value-blind differential testing misses reassociation defects.

13 LIMITATIONS

MatrixLang is straight-line and statically shaped; loops would need the facts to reach a fixed point, which the lattice permits but we have not implemented in MatrixLang. ExactReduce works on loops, but only on reductions whose terms come from integer conversions, in loops it can bound or version. The bounded contract promises the standard-model bound; with gradual underflow both orders need an absolute error term, which we do not show to be invariant. Grids of real-valued data

are 2^{-1074}, so bit-identity proofs apply to integer and dyadic data, not to general reals. Domains are trusted declarations checked at load time; a domain that is wider than the data loses opportunities, never soundness. The threshold is the true boundary for the chain families of RQ3, not in general: magnitude bounds cannot see, for instance, that every input is even. Runtime was measured on two machines with three compiler versions; other targets, notably those with ordered vector reductions, may value the license differently. The equal-cost ablation in RQ6 cannot discriminate between two mechanisms.

14 CONCLUSION

A floating-point rewrite keeps one of three guarantees, and which one is often provable from the values a program will see. Declared input domains, a grid–magnitude analysis and two theorems let a compiler reorder chains under strict semantics where every order returns the correctly rounded product, under a bounded contract where the error bound is unchanged, and report the guarantee of every output. The same facts tell a production compiler which sums it may reorder, and LLVM already has them for integer data converted where it is summed: licensing exactly those reductions gives strict C programs the speed of `-ffast-math` on them, with the strict bits, and leaves every other reduction alone. Checking certificates against execution found a real defect, and inputs built from the proof obligations killed mutants that random inputs missed.

DATA AVAILABILITY

The compilers, the LLVM pass, experiment scripts, raw observations and workspace are in the artifact, which is available to reviewers. `make contracts` repeats the MatrixLang measurements and `make contracts-analyze` regenerates their numbers, tables and figures; `tools/contracts/llvm_*.py`, `e2e_real.py` and the two further analysis scripts do the same for RQ4 on real data and on M2, and for RQ7–RQ9.

USE OF GENERATIVE AI

Generative AI tools assisted with implementation, literature search, evaluation scripts, analysis and drafting of this manuscript. The authors reviewed all text, code and results and take responsibility for them.

REFERENCES

[1] David Goldberg. 1991. What Every Computer Scientist Should Know About Floating-Point Arithmetic. *ACM Comput. Surv.* 23, 1 (1991), 5–48. https://doi.org/10.1145/103162.103163

[2] IEEE. 2019. *IEEE Standard for Floating-Point Arithmetic*. IEEE Std 754-2019.

[3] Free Software Foundation. GCC Manual: Options That Control Optimization (`-fassociative-math`, `-fno-signed-zeros`, `-fno-trapping-math`). https://gcc.gnu.org/onlinedocs/gcc/Optimize-Options.html

[4] LLVM Project. LLVM Language Reference Manual: Fast-Math Flags. https://llvm.org/docs/LangRef.html#fast-math-flags

[5] LLVM Project. Auto-Vectorization in LLVM. https://llvm.org/docs/Vectorizers.html

[6] OpenMP Architecture Review Board. 2021. *OpenMP Application Programming Interface, Version 5.2*, Section 5.5.6, Reduction Scoping Clauses. https://www.openmp.org/spec-html/5.2/openmpsu50.html

[7] Nicholas J. Higham. 2002. *Accuracy and Stability of Numerical Algorithms* (2nd ed.). SIAM.

[8] T. C. Hu and M. T. Shing. 1982. Computation of Matrix Chain Products. Part I. *SIAM J. Comput.* 11, 2 (1982), 362–373.

[9] Heiko Becker, Eva Darulova, Magnus O. Myreen, and Zachary Tatlock. 2019. Icing: Supporting Fast-Math Style Optimizations in a Verified Compiler. In *CAV 2019*, 155–173. https://doi.org/10.1007/978-3-030-25543-5_10

[10] Heiko Becker, Robert Rabe, Eva Darulova, Magnus O. Myreen, Zachary Tatlock, Ramana Kumar, Yong Kiam Tan, and Anthony Fox. 2022. Verified Compilation and Optimization of Floating-Point Programs in CakeML. In *ECOOP 2022*, LIPIcs 222, 1:1–1:28. https://doi.org/10.4230/LIPIcs.ECOOP.2022.1

[11] Nuno P. Lopes, David Menendez, Santosh Nagarakatte, and John Regehr. 2015. Provably Correct Peephole Optimizations with Alive. In *PLDI*.

[12] David Menendez, Santosh Nagarakatte, and Aarti Gupta. 2016. Alive-FP: Automated Verification of Floating Point Based Peephole Optimizations in LLVM. In *SAS*.
[13] David Menendez and Santosh Nagarakatte. 2017. Alive-Infer: Data-Driven Precondition Inference for Peephole Optimizations in LLVM. In *PLDI*.
[14] Nuno P. Lopes, Juneyoung Lee, Chung-Kil Hur, Zhengyang Liu, and John Regehr. 2021. Alive2: Bounded Translation Validation for LLVM. In *PLDI*, 65–79. https://doi.org/10.1145/3453483.3454030
[15] Sylvie Boldo, Jacques-Henri Jourdan, Xavier Leroy, and Guillaume Melquiond. 2015. Verified Compilation of Floating-Point Computations. *J. Autom. Reason.* 54, 2 (2015), 135–163.
[16] David Monniaux. 2008. The Pitfalls of Verifying Floating-Point Computations. *ACM Trans. Program. Lang. Syst.* 30, 3 (2008).
[17] W. Kahan. 1965. Further Remarks on Reducing Truncation Errors. *Commun. ACM* 8, 1 (1965), 40. https://doi.org/10.1145/363707.363723
[18] Aldy Hernandez. 2023. How We Expanded GCC Value Range Propagation for Floats. Red Hat Developer. https://developers.redhat.com/articles/2023/01/18/how-we-expanded-gcc-value-range-propagation-floats
[19] LLVM Project. 2023. `computeKnownFPClass` and `KnownFPClass`, `llvm/include/llvm/Analysis/ValueTracking.h`, release 17. https://github.com/llvm/llvm-project/blob/release/17.x/llvm/include/llvm/Analysis/ValueTracking.h
[20] LLVM Project. 2024. InstCombine `foldFBinOpOfIntCasts`, `llvm/lib/Transforms/InstCombine`, release 18. https://github.com/llvm/llvm-project/tree/release/18.x/llvm/lib/Transforms/InstCombine
[21] Eva Darulova and Viktor Kuncak. 2014. Sound Compilation of Reals. In *POPL*.
[22] Eva Darulova, Anastasiia Izycheva, Fariha Nasir, Fabian Ritter, Heiko Becker, and Robert Bastian. 2018. Daisy – Framework for Analysis and Optimization of Numerical Programs. In *TACAS*.
[23] Alexey Solovyev, Charles Jacobsen, Zvonimir Rakamarić, and Ganesh Gopalakrishnan. 2015. Rigorous Estimation of Floating-Point Round-off Errors with Symbolic Taylor Expansions. In *FM*.
[24] Pavel Panchekha, Alex Sanchez-Stern, James R. Wilcox, and Zachary Tatlock. 2015. Automatically Improving Accuracy for Floating Point Expressions. In *PLDI*.
[25] Cindy Rubio-González, Cuong Nguyen, Hong Diep Nguyen, James Demmel, William Kahan, Koushik Sen, David H. Bailey, Costin Iancu, and David Hough. 2013. Precimonious: Tuning Assistant for Floating-Point Precision. In *SC '13*. https://doi.org/10.1145/2503210.2503296
[26] Dolores Miao, Ignacio Laguna, and Cindy Rubio-González. 2024. Input Range Generation for Compiler-Induced Numerical Inconsistencies. In *ICS '24*, 201–212. https://doi.org/10.1145/3650200.3656618
[27] Katsuhisa Ozaki, Takeshi Ogita, Shin'ichi Oishi, and Siegfried M. Rump. 2012. Error-Free Transformations of Matrix Multiplication by Using Fast Routines of Matrix Multiplication and Its Applications. *Numer. Algorithms* 59, 1 (2012), 95–118. https://doi.org/10.1007/s11075-011-9478-1
[28] Hiroyuki Ootomo, Katsuhisa Ozaki, and Rio Yokota. 2024. DGEMM on Integer Matrix Multiplication Unit. *Int. J. High Perform. Comput. Appl.* 38, 4 (2024), 297–313. https://doi.org/10.1177/10943420241239588
[29] Katsuhisa Ozaki, Yuki Uchino, and Toshiyuki Imamura. 2026. Ozaki Scheme II: A GEMM-Oriented Emulation of Floating-Point Matrix Multiplication Using an Integer Modular Technique. *Int. J. High Perform. Comput. Appl.* https://doi.org/10.1177/10943420261467787
[30] Jean-Guillaume Dumas, Pascal Giorgi, and Clément Pernet. 2008. Dense Linear Algebra over Word-Size Prime Fields: the FFLAS and FFPACK Packages. *ACM Trans. Math. Softw.* 35, 3 (2008).
[31] Takeshi Ogita, Siegfried M. Rump, and Shin'ichi Oishi. 2005. Accurate Sum and Dot Product. *SIAM J. Sci. Comput.* 26, 6 (2005).
[32] James Demmel and Hong Diep Nguyen. 2013. Fast Reproducible Floating-Point Summation. In *ARITH*, 163–172.
[33] Willow Ahrens, James Demmel, and Hong Diep Nguyen. 2020. Algorithms for Efficient Reproducible Floating Point Summation. *ACM Trans. Math. Softw.* 46, 3 (2020). https://doi.org/10.1145/3389360
[34] Intel. 2025. oneMKL Developer Guide: Obtaining Numerically Reproducible Results. https://www.intel.com/content/www/us/en/docs/onemkl/developer-guide-linux/2025-2/obtaining-numerically-reproducible-results.html
[35] Peichen Xie, Yanjie Gao, Yang Wang, and Jilong Xue. 2025. Revealing Floating-Point Accumulation Orders in Software/Hardware Implementations. In *USENIX ATC '25*, 1425–1440.
[36] Mark Stephenson, Jonathan Babb, and Saman Amarasinghe. 2000. Bitwidth Analysis with Application to Silicon Compilation. In *PLDI*.
[37] Henrik Barthels, Christos Psarras, and Paolo Bientinesi. 2021. Linnea: Automatic Generation of Efficient Linear Algebra Programs. *ACM Trans. Math. Softw.* 47, 3 (2021). https://doi.org/10.1145/3446632
[38] Xuejun Yang, Yang Chen, Eric Eide, and John Regehr. 2011. Finding and Understanding Bugs in C Compilers. In *PLDI*. https://doi.org/10.1145/1993316.1993532
[39] Yue Jia and Mark Harman. 2011. An Analysis and Survey of the Development of Mutation Testing. *IEEE Trans. Softw. Eng.* 37, 5 (2011).

[40] Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Gallagher, and Tina Eliassi-Rad. 2008. Collective Classification in Network Data. *AI Magazine* 29, 3 (2008), 93–106. https://doi.org/10.1609/aimag.v29i3.2157
[41] Thomas N. Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *ICLR*.
[42] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. Gradient-Based Learning Applied to Document Recognition. *Proc. IEEE* 86, 11 (1998), 2278–2324. https://doi.org/10.1109/5.726791
[43] Christian Heimes and Brett Cannon. 2019. PEP 594 – Removing Dead Batteries from the Standard Library. https://peps.python.org/pep-0594/
[44] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. 2020. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods* 17 (2020), 261–272. https://doi.org/10.1038/s41592-019-0686-2
[45] Louis-Noël Pouchet and Tomofumi Yuki. 2016. PolyBench/C 4.2.1. https://sourceforge.net/projects/polybench/